Hanzhi "Leo" Liu

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Last updated: September 1, 2026

Education

EMORY UNIVERSITY, Atlanta, GA — Aug 2024 - May 2027 (expected)

- **Major:** Physics BS
- **GPA:** 3.97/4.0

Research Experience

EMORY UNIVERSITY, Atlanta, GA — Aug 2025 - Current
Department of Physics
Advisor: Dr. Justin Burton

- Using a lab-scale glacier analog to validate the use of physics-informed neural networks in inferring rheological properties of geophysical fluids. Honors thesis project.

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA — Summer 2025, Summer 2026
School of Earth and Atmospheric Sciences
Advisor: Dr. Winnie Chu

- Summer 2026: A deep-learning and Bayesian approach to calibrate observation data against physics-based simulations, applied to radar attenuation rate data in the Northern Greenland Ice Sheet. Publication in preparation.
- Summer 2025: Constructed an ensemble of thermal models for the Northern Greenland Ice Sheet. Developing and running numerical simulations to investigate the uncertainty in ice sheet temperatures.

Conference

AGU25, New Orleans, LA — Dec 2025

- **Liu L.**, Yang, D., Chu, W., MacGregor J. A., "*Using Thermal Model Ensemble and Machine Learning to Predict the Basal Temperature of the Northern Greenland Ice Sheet*".
- Dawson, E., Christoffersen, M., Yang, D., **Liu, L.**, Chu, W., "*Basal Temperatures from Radar Sounding: New Inferences Across East Antarctica*".

Awards & Scholarships

Emory Conference Presentation Grants — 2025

- Awarded $750 by Emory's Undergraduate Research Program for traveling to AGU25.

Emory Pathways Internship Funding — 2025

- Awarded $4000 by Emory's Pathways Center, in support of undergraduate research assistant role at Georgia Institute of Technology, summer 2025.

National Merit Scholarship — 2024

- $2000 per year scholarship. Sponsored by the National Merit Scholarship Corporation and Emory University. Selected as a National Merit Scholarship Finalist from graduating seniors across the U.S.

Extracurriculars

Emory College Honor Council, Atlanta, GA — Aug 2026 - Current

- Selected to serve on the Emory College Honor Council to enforce and promote academic integrity on campus.

AGU Cryosphere Canvassing Committee Feb 2026 - Current

- Contact and connect with AGU Cryosphere Section members, promote nominations to AGU awards.

California Science Center, Los Angeles, CA Oct 2023 - May 2024

- Educate visitors about the Space Shuttle Endeavour, and other space-related artifacts. Actively engage in educational conversations with visitors about the space science, engineering, and technology.

Skills

Programming languages: Python (PyTorch, scikit-learn, NumPy, matplotlib, pandas), MATLAB, Julia, Java.

Laboratory equipment: Rheometer, laser cutter.

Softwares: Microsoft Office Suite, 3D Modeling (Blender, Substance Painter),

Spoken languages: English (bilingual proficiency), Chinese (native proficiency), Japanese (intermediate working proficiency), French (limited working proficiency).